CYBER SECURITY HANDS-ON LAB

Task 1: VirtualBox & Kali Linux Setup Report

TASK NAME	TARGET OS	NETWORK MODE	ASSIGNED IP
Lab Setup (Task 1)	Kali Linux 2026.2	NAT Network	10.0.0.2 / 24

Verification Summary

Configuration Item	Required Parameter	Observed Status	Result
VirtualBox Network	NAT Network (10.0.0.0/24)	Configured with DHCP Enabled	VERIFIED
Adapter Mode	Attached to NatNetwork	Promiscuous: Allow All	VERIFIED
Static IP Address	10.0.0.2 / Netmask 24	Manual Configuration via NM	VERIFIED
DNS & Gateway	GW: 10.0.0.1 | DNS: 8.8.8.8	Configured & Active	VERIFIED
Host Integration	Shared Folder & Clipboard	Auto-mounted (Full Access)	VERIFIED

Setup Verification Evidence

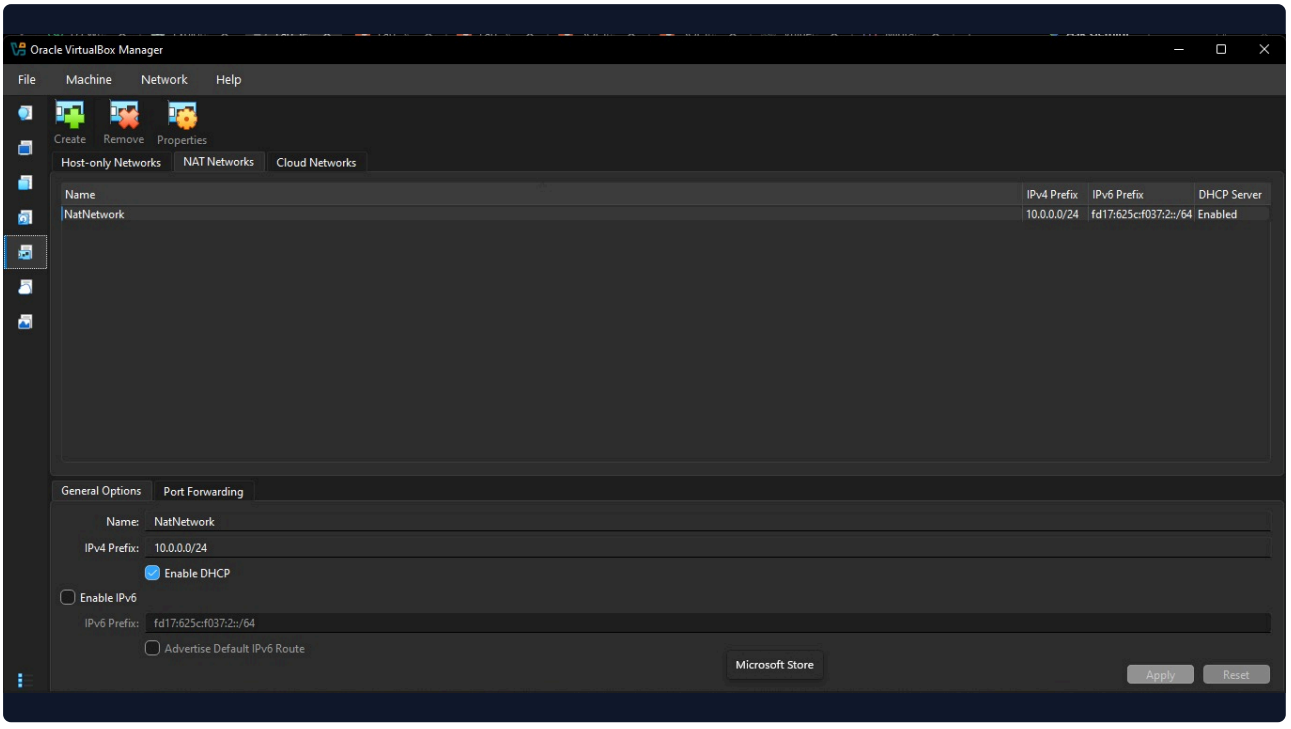


Figure 1: VirtualBox NAT Network Configuration.
Shows the global VirtualBox network tools where the custom 'NatNetwork' has been created successfully.
The IPv4 Prefix is explicitly defined as 10.0.0.0/24 with IPv4 DHCP Server enabled for internal address pool distribution.

VM Network & System Configuration

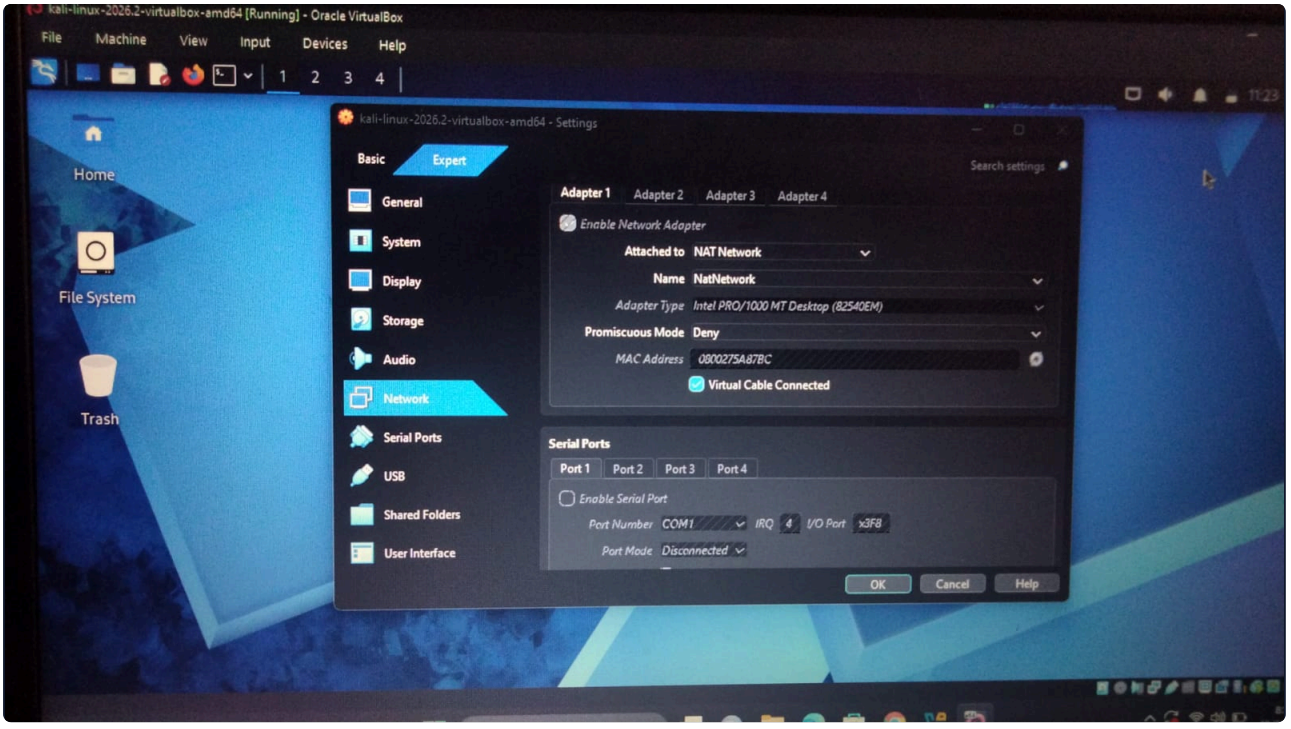


Figure 2: Virtual Machine Network Adapter Settings.
Displays Adapter 1 attached to the 'NAT Network' named 'NatNetwork' within Kali Linux VM settings. Promiscuous Mode is configured to 'Deny' in this baseline setting prior to updating to 'Allow All'.

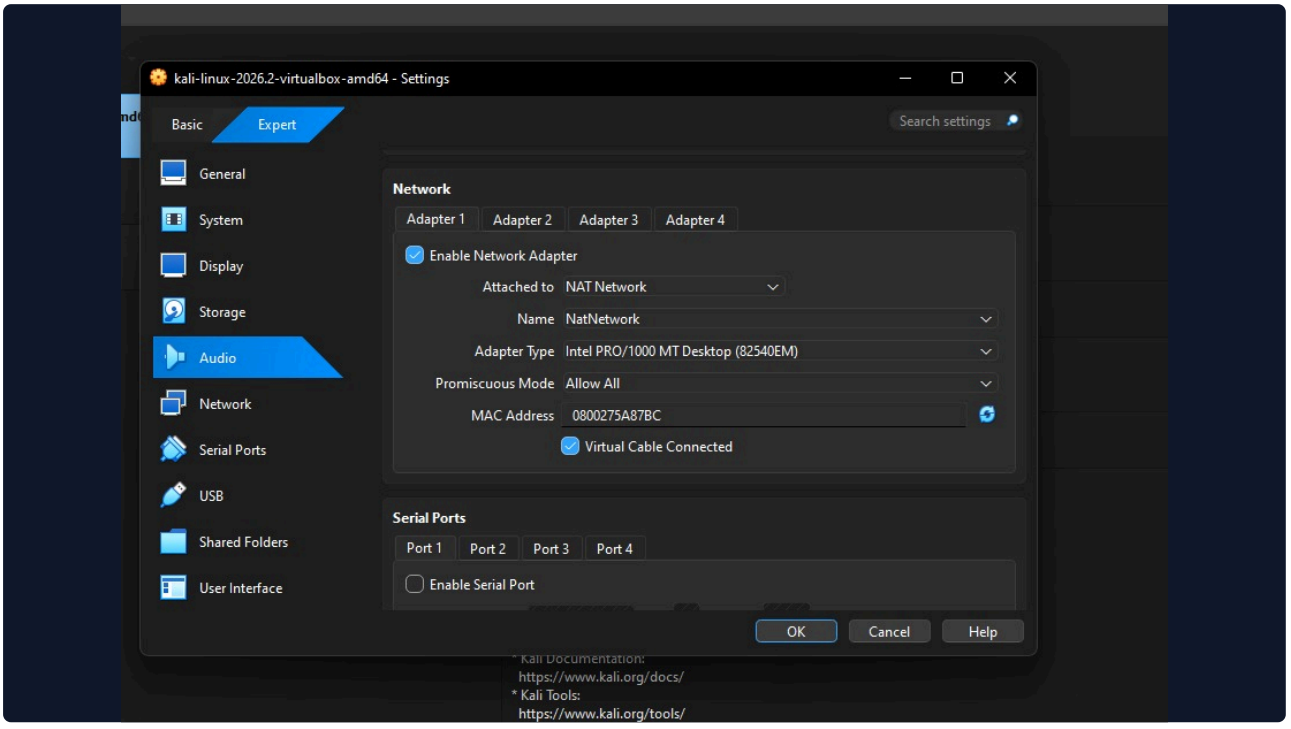


Figure 3: Advanced Promiscuous Mode Configuration.
Verifies updated Network Adapter settings with Promiscuous Mode explicitly set to 'Allow All'. This step ensures complete packet sniffing capabilities across the internal virtual lab environment.

IP Addressing & Guest Integration Settings

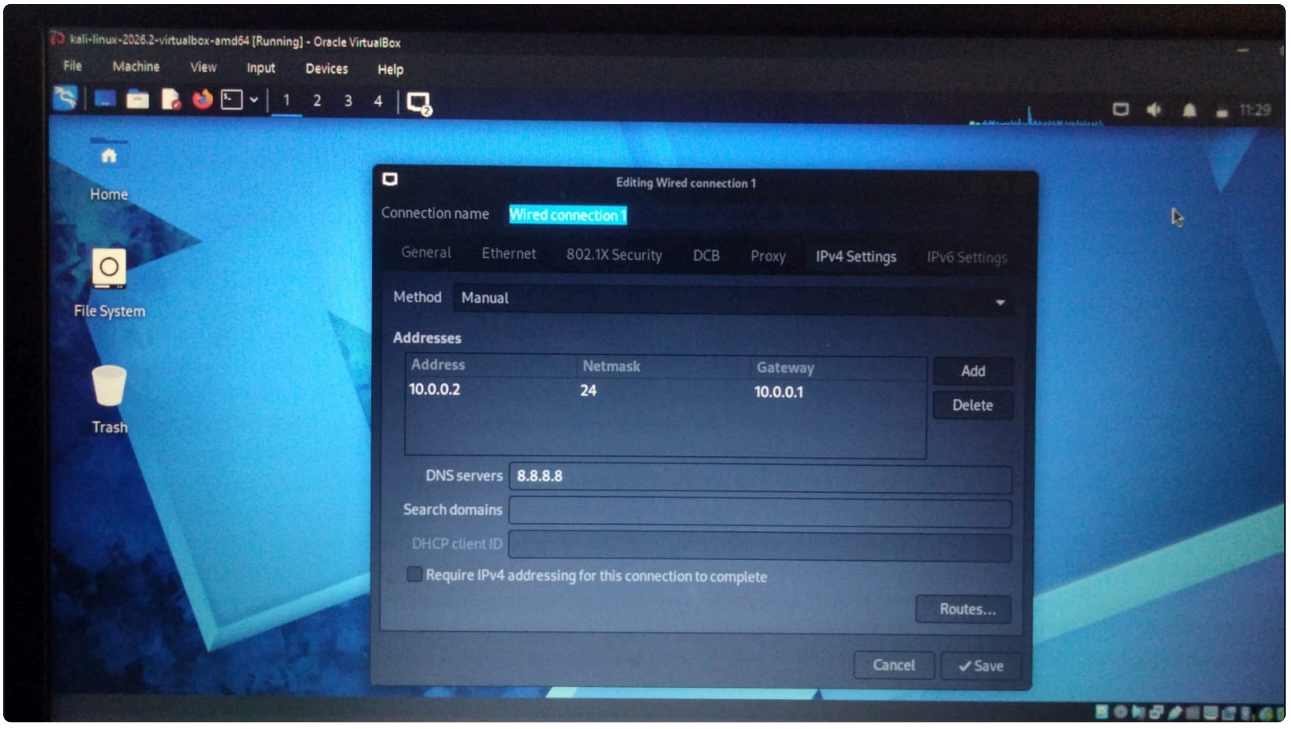


Figure 4: Manual IPv4 Address Settings in Kali GUI.
Illustrates the manual network configuration interface for 'Wired connection 1' in Kali Linux.
Static IP 10.0.0.2, Netmask 24, Gateway 10.0.0.1, and DNS server 8.8.8.8 are assigned properly.

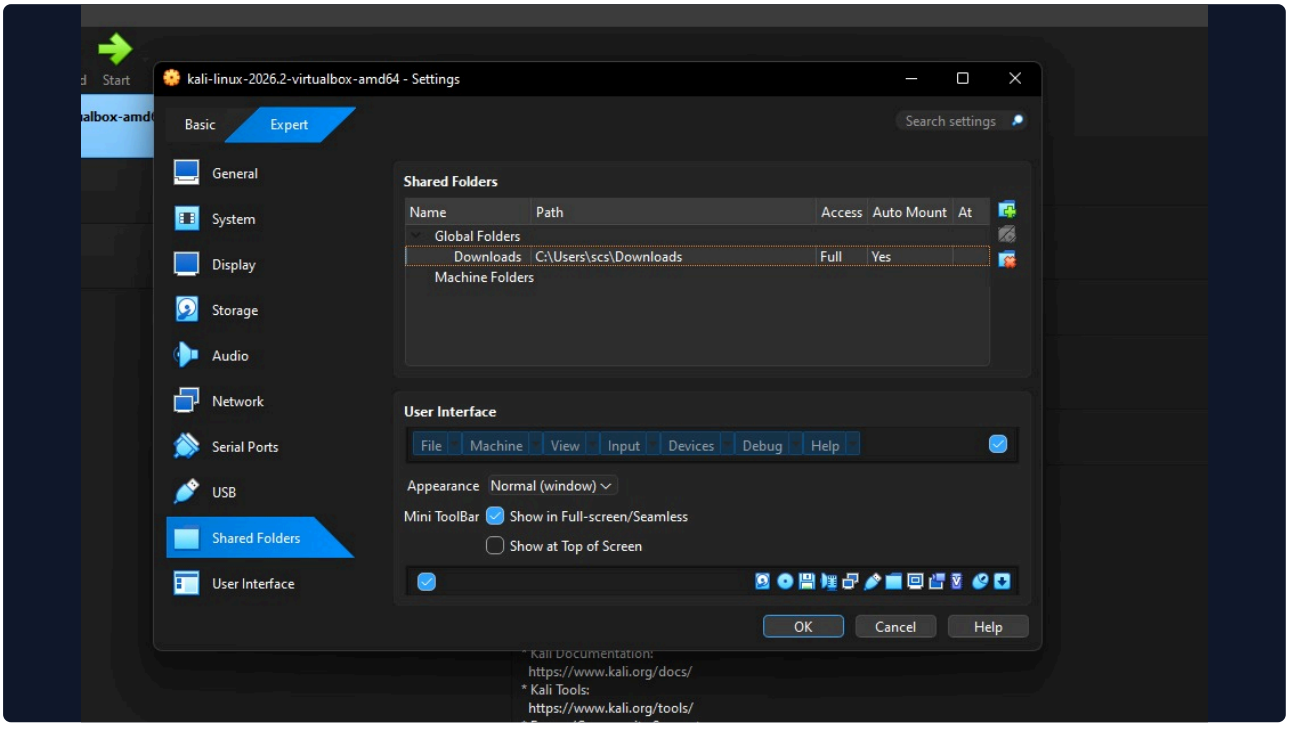


Figure 5: Host-to-Guest Shared Folder Configuration.
Confirms global shared folder setting linking host machine's 'Downloads' directory to the virtual machine.
Full access privileges and Auto-Mount options are selected to allow smooth file sharing.

Terminal Verification & System Execution



Figure 6: NetworkManager Timeout Fix Execution.
Shows successful execution of nmcli commands in terminal setting 'ipv4.dad-timeout' to 0.
The connection was successfully restarted to prevent network connection delays on VirtualBox v7.

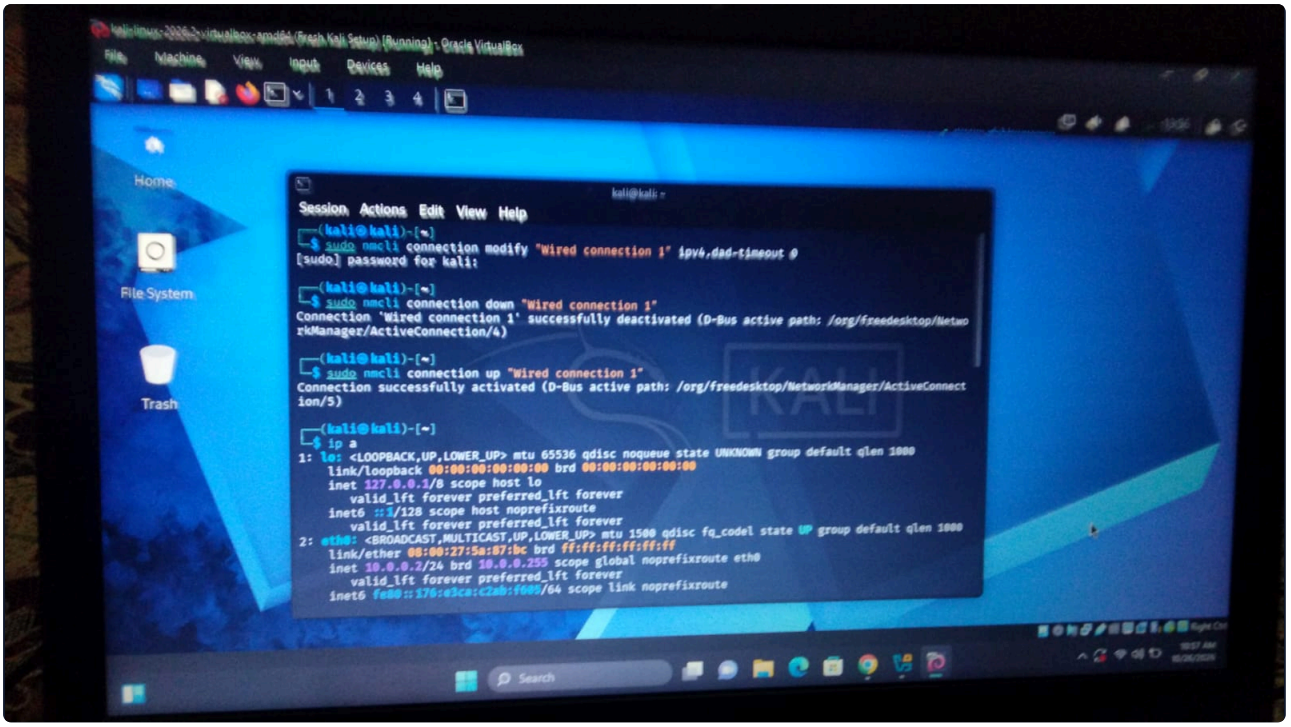


Figure 7: Final IP Verification via Terminal (`ip a`).
Displays terminal verification output of `ip a` showing adapter 'eth0' actively bound to 10.0.0.2/24.
Confirms that all IP configuration, route creation, and network manager tweaks are fully functional.